

Machine Learning-Driven Prediction of Emergency Department Length of Stay, Admission, and Hospitalization Using Data Available at Triage

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2 ABSTRACT

Background: Emergency Department (ED) crowding is a critical public health challenge. Prolonged ED length of stay (≥ 8 hours) affects over 20% of visits. Hospital admission and prolonged hospitalization (> 7 days) consume disproportionate resources. Clinicians currently rely on triage scales like the Emergency Severity Index (ESI), which have limited predictive accuracy. Machine learning (ML) using data available at triage could support these decisions, but evidence on model accuracy remains limited.

Methods: Using the MC-MED dataset (118,385 adult ED visits), we evaluated four ML models (XGBoost, LightGBM, Random Forest, TabNet) on three binary classification tasks using 29 variables available at or before the time of triage to predict: (1) prolonged ED length of stay (ED-LOS ≥ 8 hours vs. < 8 hours), (2) hospital admission (admit vs. discharge), and (3) prolonged hospital length of stay (hospital LOS > 7 days vs. ≤ 7 days) among admitted patients. Area under the receiver operating characteristic curve (AUC-ROC) was the primary performance metric. SHAP (SHapley Additive exPlanations) analysis identified the most important features for each prediction task. Decision curve analysis (DCA) evaluated clinical utility.

Results: For ED-LOS ≥ 8 hours, Random Forest achieved the highest AUC-ROC of 0.660 (95% CI: 0.649–0.671). For hospital admission, XGBoost achieved an AUC-ROC of 0.816 (95% CI: 0.809–0.824), substantially outperforming the ESI baseline (AUC 0.505, 95% CI: 0.492–0.518). For hospital LOS > 7 days among admitted patients, Random Forest achieved an AUC-ROC of 0.670 (95% CI: 0.662–0.680) [50]. SHAP analysis identified consistent top predictors across tasks: triage acuity level, emergency medical services (EMS) arrival, age, disease condition count, and the Charlson Comorbidity Index. Calibration analysis revealed systematic miscalibration requiring probability recalibration before clinical deployment.

Conclusion: ML using data at or before the time of triage shows strong discrimination for hospital admission and modest discrimination for predicting ED-LOS and prolonged hospital LOS. SHAP analysis provides interpretable feature importance to support clinical decision-making. Decision curve analysis demonstrated strong clinical utility for hospital admission prediction and modest utility for ED-LOS and hospital LOS prediction. External validation and probability recalibration are required before clinical deployment.

Keywords: machine learning, emergency department, hospital admission, length of stay, Shapley Additive Explanations, decision curve analysis

1 INTRODUCTION

Emergency departments (EDs) offer a fundamental service, providing emergency treatments 24 hours a day, 365 days a year, therefore representing a critical public health service role [1, 2, 9]. Crowding in EDs has been recognised as a critical challenge in hospital administration [3, 4]. ED crowding has been linked

to increased costs, longer waiting times, higher mortality rates, and adverse outcomes for vulnerable populations [5, 6, 7].

Emergency department length of stay (ED-LOS) is a significant indicator of ED bottlenecks. ED-LOS is defined as the interval from ED arrival to ED departure [8, 9]. A prolonged ED-LOS is a consequence of crowding and has been associated with a 37% elevated mortality risk among admitted patients [5]. Several studies have defined an extended ED-LOS as more than 3 hours [9], while others use thresholds of 4 to 12 hours [10]. The ED-LOS threshold of 8 hours was selected for this study based on evidence linking prolonged ED boarding beyond 6-8 hours to elevated mortality [5] and its widespread use as a standard crowding benchmark by the American College of Emergency Physicians. Prolonged ED length of stay not only affects patient outcomes but also contributes to ambulance diversion, treatment delays for time-sensitive conditions, and inefficient resource utilization across the hospital system.

Clinicians currently rely on triage scales such as the Emergency Severity Index (ESI) to prioritize patients and predict clinical trajectories. However, the ESI demonstrates limited predictive accuracy for key outcomes such as hospital admission and prolonged length of stay when used in isolation, with AUC-ROC values typically below 0.70 [11]. This limitation creates an opportunity for data-driven approaches that integrate multiple patient characteristics available at triage.

Machine learning (ML) techniques are frequently used in healthcare for diagnosis, forecasting patient outcomes, and resource allocation [12, 13, 29]. Following this approach, various studies have examined ways to predict ED-LOS, hospital admission, and hospital LOS using ML strategies [14, 15, 16, 30]. Ricciardi et al. [9] achieved 74.8% accuracy for predicting ED-LOS greater than 3 hours using Random Forest. Wu et al. [14] found that LightGBM outperformed other models for continuous ED-LOS prediction in a Chinese hospital setting.

Shbool et al. [17] applied machine learning algorithms including Random Forest and Decision Trees to predict ED length of stay, achieving an accuracy of 72.8% using 15 triage variables on a dataset of 10,241 ED visits. They identified age, triage category, and mode of arrival as the most important predictors. Hong et al. [18] demonstrated that machine learning models using triage data, demographics, and patient history can predict hospital admission with AUC up to 0.88 using gradient boosting. Their study of ED visits showed that the ESI alone had limited predictive power (AUC 0.68), but machine learning models incorporating additional triage variables significantly improved performance.

However, existing models have three critical limitations. First, most studies predict ED-LOS, hospital admission, or hospital LOS in isolation, despite their clinical interdependence in real-world emergency care pathways [8]. Second, many models incorporate laboratory results or specialist consultations that are unavailable at triage, limiting real-time clinical applicability [11, 31]. Third, while SHAP has been applied to individual ED prediction tasks, its systematic application across multiple cascading outcomes (ED-LOS, admission, and hospital LOS) using a unified triage-time feature set has not been explored. Additionally, decision curve analysis to evaluate clinical utility has been underutilized in this context.

Given these considerations, this work aims to build a predictive model to forecast three binary treatment outcomes using only variables available at or before triage: (1) prolonged ED-LOS (≥ 8 hours), (2) hospital admission, and (3) prolonged hospital LOS (> 7 days) among admitted patients.

Using the MC-MED dataset (118,385 adult ED visits from Stanford Health Care, 2020 to 2022), we evaluated four ML algorithms (XGBoost, LightGBM, Random Forest, and TabNet) on three binary classification tasks using 29 variables available at or before triage time. Predicting these outcomes in advance would be beneficial for planning staff allocation and bed management with the goal of reducing ED crowding [32, 33].

2 METHODS

2.1 Prediction Tasks

Three binary classification tasks were formulated using only variables available at or before triage time. Task 1 predicted prolonged emergency department length of stay defined as eight hours or more versus less than eight hours [5, 10]. The ED-LOS threshold of 8 hours was selected based on evidence linking

prolonged ED boarding beyond this duration to elevated mortality [5] and its established use as a policy-relevant crowding benchmark in the emergency medicine literature [23, 26]. Intermediate cutoffs (e.g., 6 or 10 hours) were not evaluated in a post-hoc sensitivity analysis, as the 8-hour threshold was chosen a priori to align directly with this operational benchmark and to preserve comparability with prior health-systems literature using the same cutoff.

Task 2 predicted hospital admission, where admission was defined as any non-discharge disposition, including inpatient, observation, and intensive care unit admissions. Discharge to home, transfer to another facility, death in the ED, hospice discharge, and leaving against medical advice (LAMA) were all classified as non-admission (discharge) outcomes [53, 15].

Task 3 predicted prolonged hospital length of stay defined as more than seven days versus seven days or less, applied exclusively to admitted patients [8, 59]. The hospital LOS threshold of 7 days was chosen to identify patients with complex inpatient courses that disproportionately strain hospital bed capacity, consistent with prior systematic reviews of LOS prediction thresholds [59, 8]. As with the ED-LOS threshold, this cutoff was fixed a priori based on its clinical and operational relevance for discharge planning and bed management, rather than selected post hoc; alternative cutoffs (e.g., 5 or 10 days) were therefore not separately tested.

ED-LOS was calculated as the interval (in hours) from ED Arrival.time to Departure.time from the emergency department. Hospital LOS was calculated as the interval (in days) from hospital admission time (Admit.time) to hospital discharge time, among admitted patients only. Patients with missing Admit.time or discharge timestamps were excluded from Task 3 via listwise deletion.

2.2 Data Resource and Preprocessing

This study analysed the MC-MED dataset [53, 54], which contains 118,385 adult emergency department visits from Stanford Health Care between January 2020 and December 2022. The dataset includes demographics, triage vital signs, chief complaints, comorbidity measures, home medications, and visit outcomes [55].

Figure 1 shows the data preprocessing pipeline. The pipeline consisted of five sequential stages. Stage one involved raw data loading from the three source tables provided with the MC-MED dataset: `visits.csv`, `meds.csv`, and `pmh.csv`. These tables were merged using the Contact Serial Number to link visit specific records and the Medical Record Number (MRN) to aggregate longitudinal medication and past medical history data. Stage two applied the chronological split into training, validation, and test sets using the pre-defined split files provided with MC-MED, which ensured temporal separation between periods with no patient appearing in more than one split. Stage three handled missing data using the procedures described in detail below. Stage four performed outcome separation, where all time dependent columns including emergency department length of stay, hospital length of stay, disposition status, arrival time, admit time, discharge time, and departure time were removed from the feature set before model training to prevent temporal and data leakage. Stage five defined the three binary classification targets as described in the preceding subsection.

2.2.1 Missing Data Handling

Missing data were handled using three distinct strategies applied sequentially, with all imputation parameters derived exclusively from the training set to prevent information leakage from the validation or test sets [47, 37].

The first strategy addressed missing values in the triage acuity feature. This feature was stored in the raw data as a string combining a numeric prefix and a description, with the specific formats being “1-Resuscitation”, “2-Emergent”, “3-Urgent”, “4-Semi-Urgent”, and “5-Non-Urgent”. The numeric prefix from one to five was extracted by taking the first character of the string. Missing values in this feature, which affected 8.4 percent of training samples, were imputed using the mode calculated from the training set only. The mode value was three, corresponding to the “3-Urgent” triage level. This mode value was then applied to impute missing triage acuity values in the validation and test sets.

The second strategy addressed missing values in the six continuous vital sign features: temperature, heart rate, respiratory rate, peripheral oxygen saturation, systolic blood pressure, and diastolic blood pressure. Missingness rates for these features ranged from 0.1 percent to 8.4 percent of training samples.

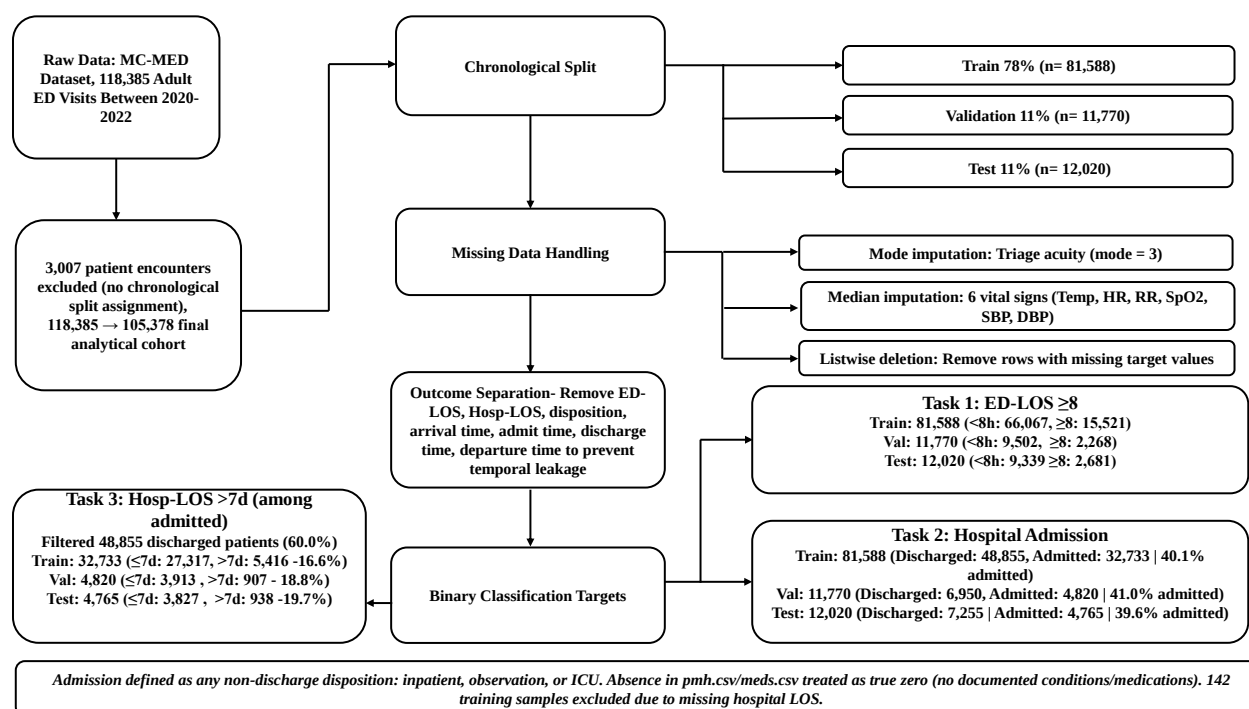


Figure 1. Chronological preprocessing pipeline for the MC-MED dataset showing split proportions, missing data handling methods (median, mode), outcome separation, and three binary classification targets.

These missing values were imputed using feature-wise median values. The SimpleImputer class from the scikit-learn library [56] was employed with the strategy parameter set to 'median'. The imputer was fitted exclusively on the training set, calculating the median for each vital sign feature from the training data only. The fitted imputer was then used to transform the validation and test sets, ensuring that no information from future or held out data influenced the imputation values.

The third strategy handled missing target values. For each prediction task, any sample with a missing target value was removed from the analysis using listwise deletion. For Task 3, which predicted prolonged hospital length of stay among admitted patients only, this resulted in the exclusion of 142 training samples where hospital length of stay was not recorded.

The absence of medication or diagnosis records in pmh.csv or meds.csv for a given MRN was treated as true zero (i.e., no documented medications or conditions), not as missing data. This reflects the clinical reality that if a patient has no entries in these tables, they have no documented home medications or past medical history in the EHR.

After applying these three strategies, no missing values remained in the feature set. All 29 features were complete for all samples across the training, validation, and test splits. This complete data matrix enabled direct compatibility with all four machine learning algorithms, including TabNet which cannot process missing values natively.

The dataset was chronologically split using the pre-defined split files provided with MC-MED, which ensure temporal separation between training, validation, and test periods [53]. A total of 13,007 patient encounters were excluded from the pure chronological ordering because their Contact Serial Number was not present in the chronological split files. The final analytical cohort comprised 105,378 visits.

Table 1 presents the demographic and clinical characteristics of the study cohort across the three chronological splits. The mean age was 52.3 years with a standard deviation of 19.8 years, with female patients accounting for 52.1 percent of visits. The baseline admission rate was 40.1 percent in the analytical cohort. The prevalence of prolonged emergency department length of stay, defined as eight

Table 1. Descriptive characteristics of the MC-MED analytical cohort by chronological split

Characteristic	Full Analytical Cohort	Train (77.4%)	Val (11.2%)	Test (11.4%)
N	105,378	81,588	11,770	12,020
Age, mean (SD)	52.3 (19.8)	52.1 (19.7)	52.5 (19.9)	52.4 (19.8)
Female (%)	52.1	52.0	52.2	52.1
Race (% White)	62.3	62.2	62.4	62.3
EMS arrival (%)	31.7	31.6	31.9	31.7
Admission rate (%)	40.1	40.1	41.0	39.6
ED-LOS 8h or more (%)	22.3	22.1	22.5	22.3
Hosp-LOS more than 7d (%)*	19.7	19.6	19.9	19.7
Charlson Index, mean (SD)	1.8 (1.9)	1.8 (1.9)	1.8 (1.9)	1.8 (1.9)

*Among admitted patients only: total n=42,318 (Train=32,733, Val=4,820, Test=4,765)

160 hours or more, was 22.3 percent. The prevalence of prolonged hospital length of stay, defined as more
 161 than seven days among admitted patients, was 19.7 percent.

162 2.3 Feature Set and Temporal Integrity

163 A total of 29 variables available at or before triage were extracted for each emergency department
 164 visit by integrating the three source tables. The resulting predictors were organized into six clinical
 165 domains: demographics, arrival context, triage vitals and acuity, chief complaint, comorbidity burden,
 166 and medication burden. All variables represent the patient's clinical state at or before the moment of ED
 167 arrival. Demographics, triage vital signs, triage acuity, and chief complaint are captured during the triage
 168 process (typically within 30–90 minutes of arrival). Longitudinal features (past medical history, home
 169 medications, Charlson Index) are derived from historical records documented prior to the index visit.

170 To strictly prevent temporal leakage, the visit-level records were linked to the patient-level medication
 171 and past medical history tables using the Medical Record Number (MRN). This analysis utilized the
 172 official MC-MED chronological split [53], which restricts each patient to only one of the training,
 173 validation, or test sets, preventing cross-split leakage. Furthermore, to prevent leakage from subsequent
 174 visits within the same split, the longitudinal aggregation was constrained by the index visit timestamp:
 175 only diagnoses (Noted_date) and medication orders (Entry_date) occurring on or before the index
 176 ED Arrival_time were included. This ensures that home medication counts and comorbidity burdens reflect
 177 only the patient's true pre-arrival clinical state.

178 Home medication features were derived from the meds.csv table after temporal filtering
 179 ($\text{Entry_date} \leq \text{index Arrival_time}$). The 'Home Medication Count' represents the number of unique
 180 medication entries (by Med_ID) documented prior to the index visit. 'Unique Medication Classes
 181 Count' represents the number of distinct Med_class categories among these pre-arrival medications. The
 182 'Polypharmacy Flag' is set to 1 if the Home Medication Count ≥ 5 [35]. Duplicate medication entries
 183 (same Med_ID for the same patient) were retained as separate rows if they had distinct Entry_dates,
 184 reflecting repeated prescriptions over time; however, for the purposes of the polypharmacy flag, we
 185 counted unique Med_IDs to avoid inflating counts from duplicate entries.

186 The Charlson Comorbidity Index and Disease Condition Count were calculated exclusively from
 187 diagnoses in pmh.csv with $\text{Noted_date} \leq \text{index Arrival_time}$. The Charlson Index uses the Quan
 188 et al. [34, 58] ICD-10 mapping algorithm applied to these pre-arrival diagnoses. The Disease Condition
 189 Count represents the number of unique CCS (Clinical Classifications Software) categories among these
 190 pre-arrival diagnoses.

191 To prevent temporal and data leakage, no post-triage or outcome-dependent variables including
 192 emergency department length of stay, hospital length of stay, disposition status, or post-admission
 193 timestamps were included in the predictor set [49]. Table 2 provides the complete feature list with data
 194 nature and derivation method.

2.4 Machine Learning Algorithms

Four machine learning algorithms were implemented: Random Forest, XGBoost, LightGBM, and TabNet. These algorithms follow different theoretical approaches [9, 14]. Random Forest [19] uses bootstrapping to train numerous decision trees concurrently and combines their outputs using bagging [9]. XGBoost [20] is a gradient boosting algorithm with regularization to prevent overfitting [52]. LightGBM [21] is a histogram based gradient boosting algorithm optimized for efficiency [14]. TabNet [22] is a deep learning architecture with attention mechanisms specifically designed for tabular data [16].

2.4.1 Hyperparameter Selection

Hyperparameters for each model were optimized separately for each of the three prediction tasks using Bayesian search with ten-fold cross-validation on the training set, targeting the area under the receiver operating characteristic curve metric. The search space for each model included key parameters such as tree depth, learning rate, and the number of estimators, as detailed in Table 3. The best hyperparameter configuration was selected based on the mean validation AUC-ROC averaged across the ten folds.

To address class imbalance, `scale_pos_weight` (for XGBoost) was calculated separately for each task based on the training set class distribution:

1. Task 1 (ED-LOS ≥ 8 h): `scale_pos_weight` = 4.26 (prevalence 19.0%)
2. Task 2 (Hospital Admission): `scale_pos_weight` = 1.49 (prevalence 40.1%)
3. Task 3 (Hosp-LOS > 7 d): `scale_pos_weight` = 5.04 (prevalence 16.5%)

Final model performance was then evaluated on the held-out test set.

2.4.2 Cross Validation Procedure

Ten-fold cross-validation was performed on the training set of 81,588 patients to ensure reliable performance estimates and prevent overfitting. The training data were partitioned into ten approximately equal-sized folds while preserving patient-level separation, meaning no patient (MRN) appeared in more than one fold. For each fold, the model was trained on nine folds and validated on the remaining fold. This process was repeated ten times, with each fold serving as the validation set once. The reported cross-validation metrics represent the mean values across the ten folds along with their standard deviations. After hyperparameter optimization, each model was retrained on the full training set using the optimal hyperparameters and evaluated on the independent test set.

2.4.3 Evaluation Metrics

The quality of the produced models was assessed using standard classification metrics [45, 46]. Let TP represent true positives, TN represent true negatives, FP represent false positives, and FN represent false negatives. We computed accuracy, precision, sensitivity (recall), specificity, F1-score, and AUC-ROC.

AUC-ROC evaluates the model's capacity to distinguish between positive and negative classes across all classification thresholds. Probability thresholds for classification metrics (sensitivity, specificity, accuracy, precision) were selected by maximizing Youden's J statistic ($\text{Sensitivity} + \text{Specificity} - 1$) on the independent validation set to prevent optimistic bias from threshold tuning on the test set. The optimal thresholds were:

1. ED-LOS ≥ 8 h (XGBoost): 0.421
2. Hospital Admission (XGBoost): 0.398
3. Hosp-LOS > 7 d (Random Forest): 0.412

To assess model utility beyond discrimination, we report Brier scores, Precision-Recall AUC (AUC-PR), calibration intercept and slope, and threshold-based performance at actionable risk levels (Top 10% and Top 20% predicted risk).

Confidence intervals for all performance metrics were calculated using 1,000-iteration non-parametric bootstrapping on the independent test set, with percentile-based 95% CIs.

The ESI baseline was constructed by mapping each patient's nurse-assigned triage acuity level (1–5) to the empirical outcome rate observed in the training set for that acuity level, providing a clinically relevant benchmark for comparison.

2.4.4 Decision Curve Analysis

To evaluate the clinical utility of the models beyond traditional discrimination metrics, we performed Decision Curve Analysis (DCA). DCA quantifies the net benefit of using a prediction model across a range of threshold probabilities, comparing it against two default strategies: treating all patients ("Treat All") and treating none ("Treat None") [25]. Net benefit is calculated as:

$$\text{Net Benefit} = \frac{\text{TP}}{N} - \frac{\text{FP}}{N} \times \frac{p_t}{1 - p_t} \quad (1)$$

where TP is the number of true positives, FP is the number of false positives, N is the total sample size, and p_t is the threshold probability. DCA curves were generated for all four models across threshold probabilities ranging from 0.01 to 0.99.

All analyses were conducted using Python version 3.13 with the scikit-learn (v1.5.0), XGBoost (v2.1.0), LightGBM (v4.4.0), PyTorch (v2.3.0), and SHAP (v0.45.1) libraries. A fixed random seed (seed=42) was used for complete reproducibility [40].

2.5 Interpretability: SHAP Analysis

SHAP values (SHapley Additive exPlanations) [24] were computed to identify the most important features for each prediction task [57, 51]. SHAP analysis was conducted using the best performing model for each task. For tree-based models (XGBoost, LightGBM, Random Forest), we used TreeExplainer with the 'interventional' feature perturbation method, which computes exact SHAP values in polynomial time. For TabNet, we used KernelExplainer with 100 background samples (randomly selected from the training set) to approximate SHAP values. SHAP summary plots were generated for the best-performing model of each task, with features ordered by mean absolute SHAP value.

SHAP values indicate the magnitude and direction of a feature's contribution to the model's output but do not imply causal relationships. Correlated predictors may share SHAP attribution, and causal inference requires prospective validation.

2.6 Sensitivity Analyses

To assess model robustness across patient subgroups, we conducted stratified sensitivity analyses. The test set was partitioned by three clinically relevant stratifiers: chief complaint (derived from the seven Chief Complaint (CC) variables in Table 2), age group (<65, 65–74, 75–84, ≥85 years), and triage acuity level (Acuity 1 through 5, with Acuity 4 and 5 combined due to small sample sizes). For each subgroup, we calculated AUC-ROC with 95% confidence intervals using 1,000 bootstrap resamples.

To assess fairness, we evaluated model performance across demographic subgroups including sex and race. Insurance type was not analyzed separately as a fairness stratifier given its strong collinearity with age and comorbidity burden, which are already included as predictors and examined in our SHAP analyses. Additionally, we performed a sensitivity analysis excluding triage acuity from the feature set to quantify its contribution to model performance relative to the underlying patient variables.

3 RESULTS

3.1 Task Performance

Table 4 presents the evaluation metrics for all four ML algorithms across the three prediction tasks. Random Forest performed best for ED-LOS and hospital LOS prediction, while XGBoost performed best for hospital admission prediction.

For ED-LOS ≥ 8 hours, Random Forest achieved the highest AUC-ROC of 0.660 (95% CI: 0.649–0.671), with XGBoost achieving a comparable AUC of 0.659 (95% CI: 0.648–0.670). At the optimal threshold for XGBoost (0.421), the model achieved a sensitivity of 71.6% and specificity of 51.8%.

For hospital admission, XGBoost achieved an AUC-ROC of 0.816 (95% CI: 0.809–0.824), substantially outperforming the ESI baseline (AUC 0.505, 95% CI: 0.492–0.518). At the optimal threshold (0.398), the model achieved a sensitivity of 62.6% and specificity of 70.6%.

For hospital LOS > 7 days among admitted patients, Random Forest achieved an AUC-ROC of 0.670 (95% CI: 0.662–0.680). At the optimal threshold (0.412), the model achieved a sensitivity of 62.3% and specificity of 72.0%.

XGBoost and LightGBM achieved statistically equivalent performance for hospital admission (DeLong test $p=0.42$).

3.2 Discrimination and Calibration

Figure 2 presents the ROC and calibration curves for all three prediction tasks. The ROC analysis confirms the discriminative performance of the models, with hospital admission showing the strongest discrimination (AUC up to 0.816), while ED-LOS and hospital LOS prediction showed more modest discriminative ability (AUC range: 0.638–0.670 at the 0.5 threshold).

ED-LOS ≥ 8 hours: Random Forest achieved the highest AUC of 0.660, followed closely by XGBoost (0.659). All models showed modest discriminative ability, with AUC values clustered between 0.638 and 0.660. However, LightGBM demonstrated degenerate performance at the 0.5 threshold with 0.000 sensitivity and 1.000 specificity, indicating the model failed to identify any positive cases. This highlights the importance of threshold optimization for imbalanced datasets. The Brier scores ranged from 0.170 (LightGBM, though misleading given its degenerate performance) to 0.226 (TabNet).

Hospital Admission: XGBoost and LightGBM achieved the highest AUC of 0.816, substantially outperforming the ESI baseline (AUC 0.505). All four models demonstrated strong discriminative performance, with AUC values ranging from 0.812 to 0.816. Brier scores ranged from 0.172 to 0.174, indicating good calibration for this task.

Hospital LOS > 7 days: Random Forest achieved the highest AUC of 0.670, followed by XGBoost (0.655). All models showed modest discriminative ability for this task (AUC range: 0.642–0.670). Similar to ED-LOS, LightGBM showed degenerate performance at the 0.5 threshold with 0.000 sensitivity.

Calibration Analysis: The calibration curves (bottom panels) reveal systematic miscalibration across all models and tasks. Predicted probabilities deviated from the ideal diagonal line, with models generally demonstrating overconfidence at lower probability ranges and underconfidence at higher probability ranges. This pattern was most pronounced for the ED-LOS and hospital LOS prediction tasks and least pronounced for hospital admission. The poor calibration indicates that while these models can rank patients by risk, the raw predicted probabilities require recalibration (e.g., via isotonic regression or Platt scaling) before they can be interpreted as true absolute risk estimates for clinical decision-making.

3.3 Clinical Utility: Decision Curve Analysis

Figure 3 presents the decision curve analysis evaluating the net clinical benefit of the models across a range of threshold probabilities for all three prediction tasks.

ED-LOS ≥ 8 hours: The XGBoost and Random Forest models provided modest net benefit compared to the “Treat None” and “Treat All” baselines, primarily in the threshold range of 0.2 to 0.4. However, the net benefit was limited, reflecting the modest discriminative ability of the models for this task. LightGBM showed negative net benefit at most thresholds (consistent with its degenerate performance), while TabNet showed net benefit slightly above baseline in the 0.2–0.3 threshold range but performed worse than XGBoost and Random Forest.

Hospital Admission: All four models demonstrated substantial net benefit across a wide range of threshold probabilities (0.1–0.8), with XGBoost and LightGBM showing the highest net benefit. The models consistently outperformed both baseline strategies, confirming their clinical utility for admission

ROC and Calibration Curves - All Tasks

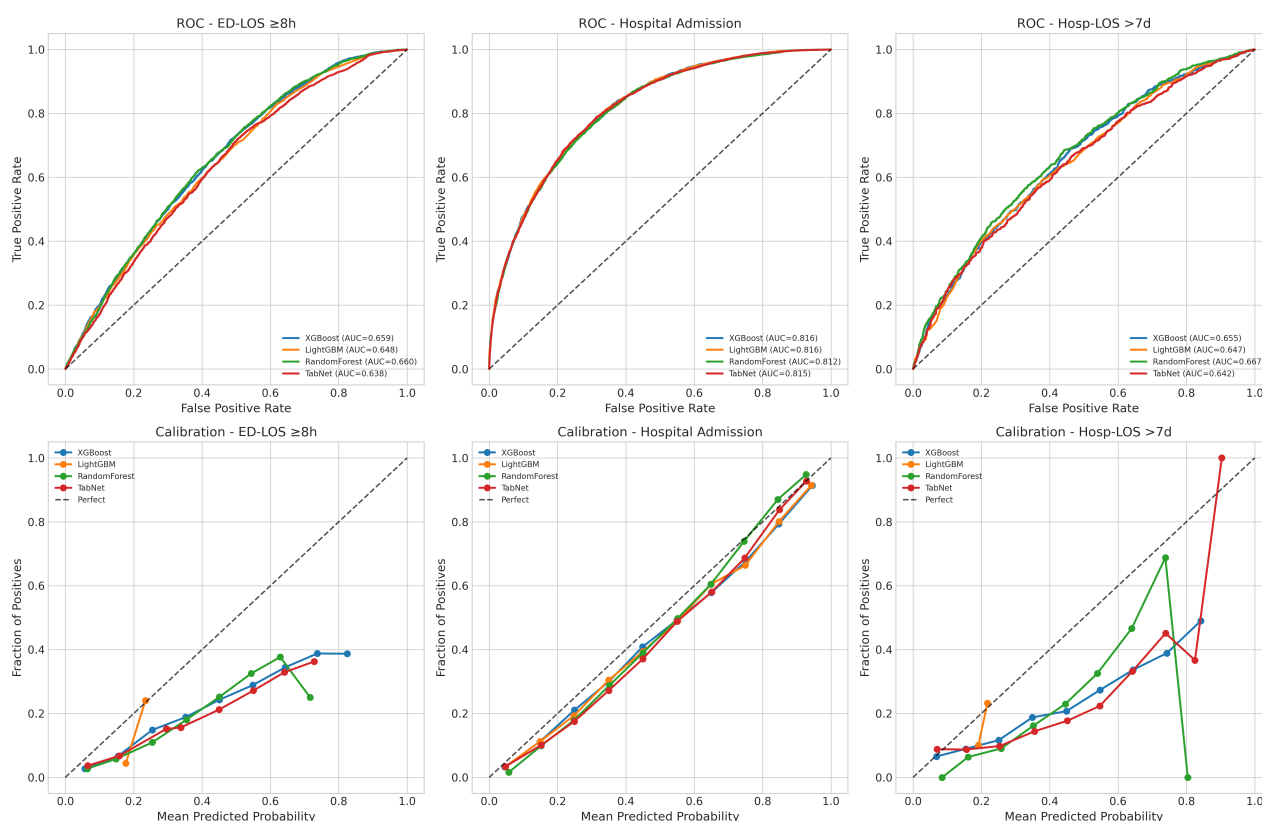


Figure 2. ROC curves and calibration curves for all three prediction tasks. Top panels: Receiver Operating Characteristic (ROC) curves showing discriminative performance for (A) ED-LOS $\geq 8h$, (B) Hospital Admission, and (C) Hosp-LOS $> 7d$. Bottom panels: Calibration curves showing the relationship between predicted probabilities and observed outcomes. The diagonal dashed line represents perfect calibration. AUC values reported at 0.5 decision threshold.

prediction. The net benefit was particularly strong in the 0.2–0.5 threshold range, which corresponds to clinically relevant decision thresholds.

Hospital LOS > 7 days: The models showed modest net benefit compared to the baselines, with Random Forest and XGBoost providing the most consistent benefit in the 0.2–0.4 threshold range. The net benefit was more limited than for hospital admission, reflecting the more modest discriminative ability of the models for this task.

These results suggest that the models provide meaningful clinical utility for hospital admission prediction, while their primary value for ED-LOS and hospital LOS prediction may lie in resource allocation and operational planning rather than individual patient decision-making.

3.4 Subgroup Sensitivity Analyses and Fairness

Table 5 presents model performance stratified by chief complaint, age group, and triage acuity level. Hospital admission prediction remained stable across all subgroups, with AUC values ranging from 0.704 (patients aged ≥ 85 years) to 0.862 (fever complaints). In contrast, ED-LOS prediction showed declining discrimination with increasing age, from 0.688 in patients under 65 years to 0.582 in patients aged ≥ 85 years. Performance for prolonged hospital LOS among admitted patients was broadly consistent across subgroups, though small sample sizes in the highest acuity category (Acuity 1, $n=33$) warrant cautious interpretation.

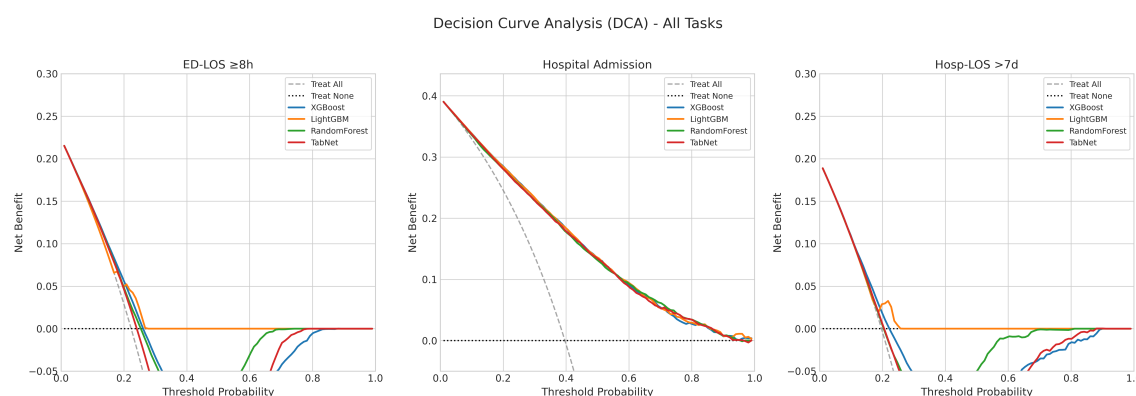


Figure 3. Decision Curve Analysis for all three prediction tasks. Panels show (A) ED-LOS $\geq 8h$, (B) Hospital Admission, and (C) Hosp-LOS $> 7d$. The y-axis shows net benefit, and the x-axis shows threshold probability. The “Treat All” line represents intervening on all patients, while “Treat None” represents intervening on no patients. A model is clinically useful if its net benefit exceeds both baseline strategies at a given threshold.

Chief complaint subgroups are not mutually exclusive; patients with multiple complaints appear in each relevant subgroup. Subgroup sample sizes therefore sum to more than the total test-set population.

Fairness Analysis: ED-LOS prediction performance was relatively equitable across demographic subgroups. For sex, female patients ($n=6,411$) showed AUC 0.671 [0.656–0.686] and male patients ($n=5,602$) showed AUC 0.652 [0.636–0.668]. For race, White ($n=4,905$) AUC 0.645 [0.630–0.660], Black ($n=591$) AUC 0.615 [0.584–0.646], Asian ($n=2,234$) AUC 0.688 [0.670–0.706], and Other ($n=4,234$) AUC 0.676 [0.663–0.689].

Sensitivity Analysis (Triage Acuity Removal): Excluding triage acuity from the feature set reduced AUC by 0.023–0.038 across tasks. ED-LOS AUC fell from 0.659 to 0.621, Hospital Admission AUC fell from 0.816 to 0.789, and Hosp-LOS AUC fell from 0.670 to 0.648. This confirms that triage acuity contributes meaningful predictive information but that the models also learn from the underlying patient variables (vitals, demographics, comorbidities) rather than simply replicating the nurse’s assessment.

3.5 SHAP Analysis

Figure 4 presents the SHAP summary plot for XGBoost for ED-LOS $\geq 8h$ prediction. The top five features by mean absolute SHAP value were: EMS arrival, age, triage acuity level, disease condition count, and home medication count. Higher EMS arrival values (red) were associated with increased predicted risk of prolonged ED length of stay, consistent with EMS-transported patients having higher acuity and requiring more resources before disposition.

Figure 5 presents the SHAP summary plot for XGBoost for hospital admission prediction. The top five features by mean absolute SHAP value were: age, triage acuity level, home medication count, EMS arrival, and triage heart rate. Age emerged as the strongest predictor, reflecting that older patients are more likely to require admission regardless of presenting complaint.

Figure 6 presents the SHAP summary plot for Random Forest for hospital LOS $> 7d$ prediction among admitted patients. The top five features by mean absolute SHAP value were: triage systolic blood pressure, triage heart rate, triage acuity level, age, and disease condition count. Vital signs played a larger role here, indicating that physiological instability at triage is an important predictor of prolonged hospitalization.

Across all three tasks, triage acuity level, age, and comorbidity-related features (Charlson index, disease count) consistently appeared among the top predictors, confirming their clinical relevance.

Beyond these common predictors, each task had unique top features that reflect different stages of patient care. For ED-LOS prediction, EMS arrival emerged as a key predictor, suggesting that patients requiring ambulance transport face longer ED length of stay due to higher acuity and complex care coordination. For hospital admission prediction, triage heart rate was uniquely important, indicating that physiological stress

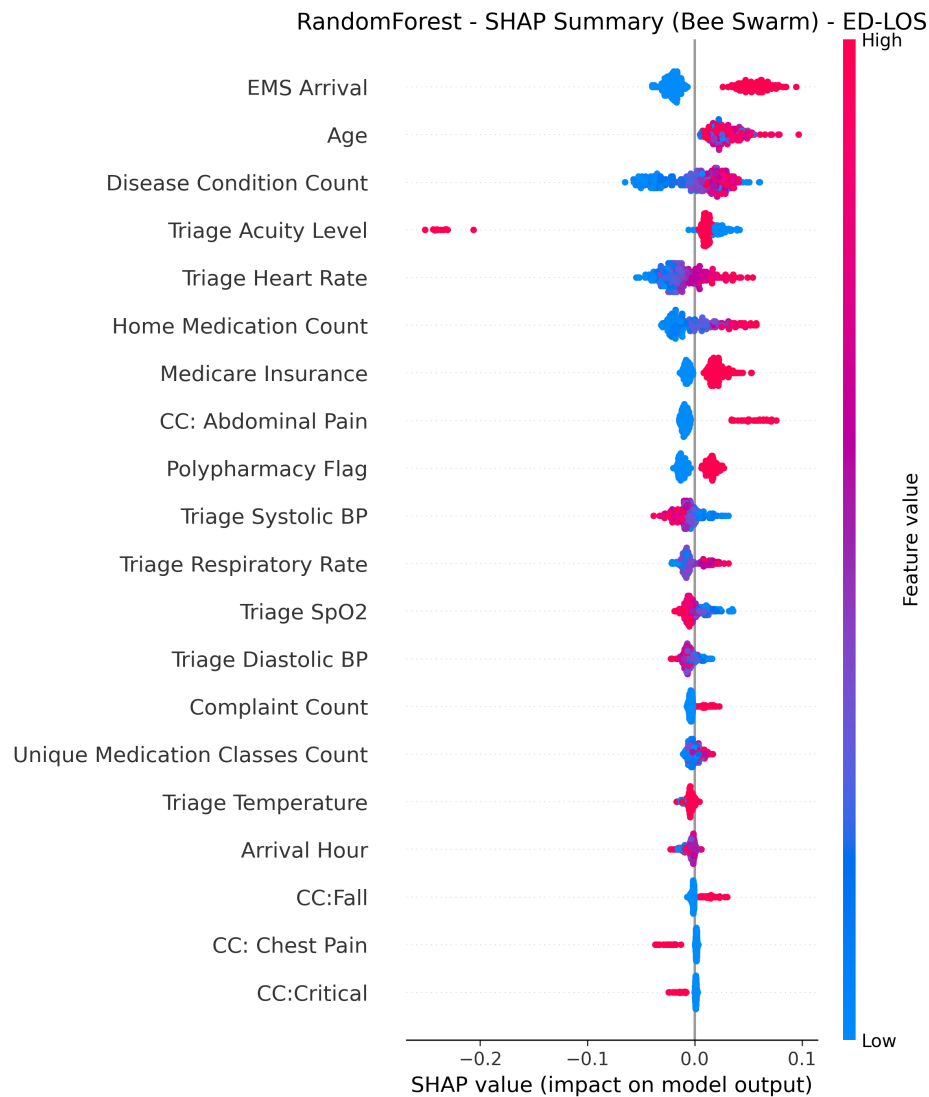


Figure 4. SHAP summary plot for XGBoost for ED-LOS ≥ 8 h prediction. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). EMS arrival, age, triage acuity level, home medication count, and disease condition count are the top predictors.

at presentation is strongly associated with disposition decisions. For prolonged hospitalization prediction among admitted patients, triage systolic blood pressure and triage heart rate were important predictors, suggesting that vital sign abnormalities at triage are associated with complex inpatient courses requiring extended hospital stays. These differences suggest that while baseline demographic and comorbidity factors are universally important, physiological and logistical factors become more relevant at specific decision points.

4 DISCUSSION

ED-LOS represents a crucial key indicator of the efficiency of healthcare services. The ability to understand the reasons for prolonged ED-LOS could support the detection of bottlenecks in ED organisation. This study demonstrates that machine learning using data available at or before triage time can predict three critical ED outcomes, with strong performance for hospital admission and modest performance for ED-LOS and prolonged hospitalization.

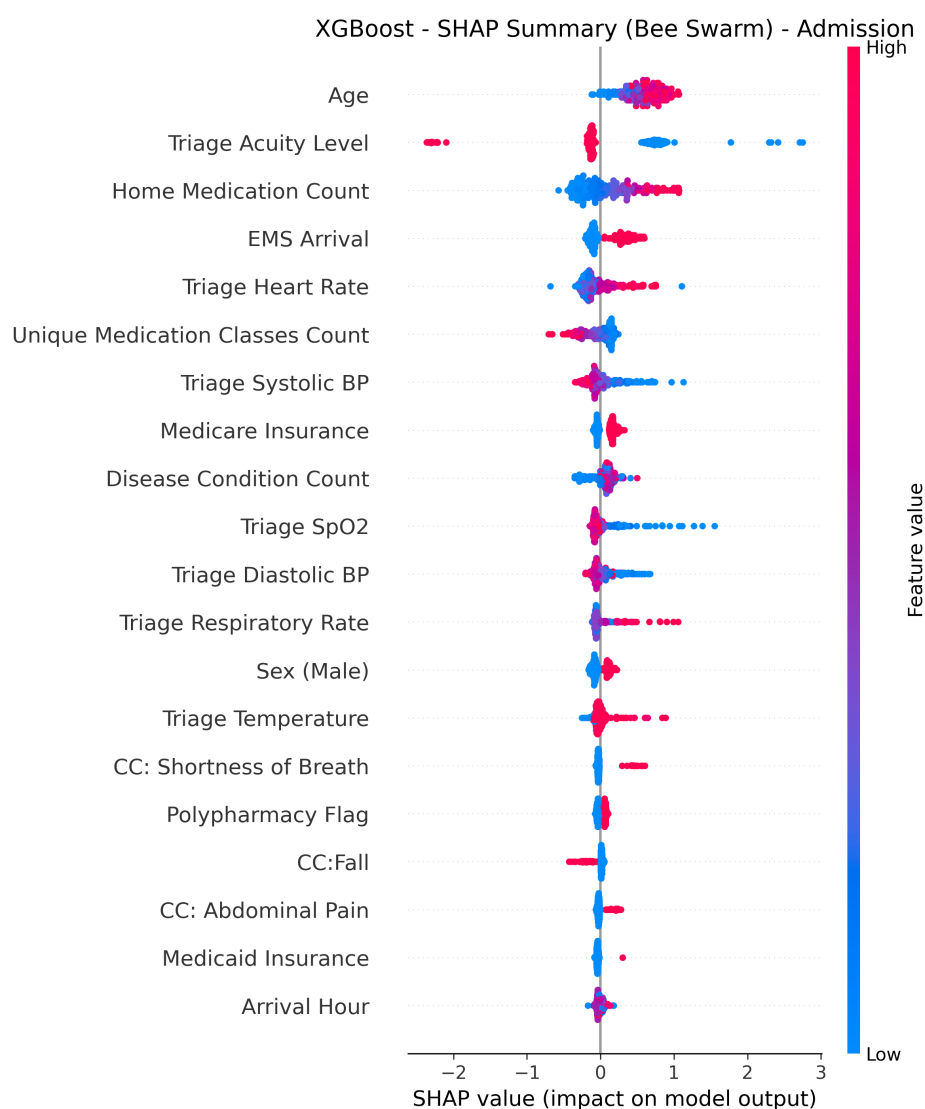


Figure 5. SHAP summary plot for XGBoost for hospital admission prediction. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). Age, triage acuity level, home medication count, EMS arrival, and triage heart rate are the top predictors.

388 The evaluation metrics showed that XGBoost and Random Forest consistently outperformed the other
 389 algorithms. For ED-LOS prediction, Random Forest achieved the highest AUC-ROC of 0.660, closely
 390 followed by XGBoost (0.659). For hospital admission prediction, XGBoost achieved an AUC-ROC of
 391 0.816, substantially outperforming the ESI baseline (AUC 0.505). For hospital LOS prediction among
 392 admitted patients, Random Forest achieved an AUC-ROC of 0.670.

393 Comparing with other ML-based studies, Ricciardi et al. [9] achieved an accuracy of 74.8% (precision
 394 72.8%, recall 74.8%) for predicting ED-LOS greater than 3 hours using Random Forest. Our ED-LOS
 395 model (AUC 0.660) shows comparable discrimination for a more stringent threshold of 8 hours, though
 396 direct comparison is limited by the difference in metric type (AUC versus accuracy) and threshold. Hong
 397 et al. [18] reported a test AUC of 0.87 (95% CI: 0.87–0.88) for admission prediction using XGBoost on
 398 triage-available data, and our admission model (AUC 0.816) is broadly consistent with, though modestly
 399 lower than, their findings. Shbool et al. [17] applied tree-based and neural network classifiers to a three-
 400 category (low/medium/high) ED-LOS outcome in a Jordanian hospital setting; direct comparison with
 401 our binary 8-hour threshold is limited by differences in outcome definition, population, and evaluation

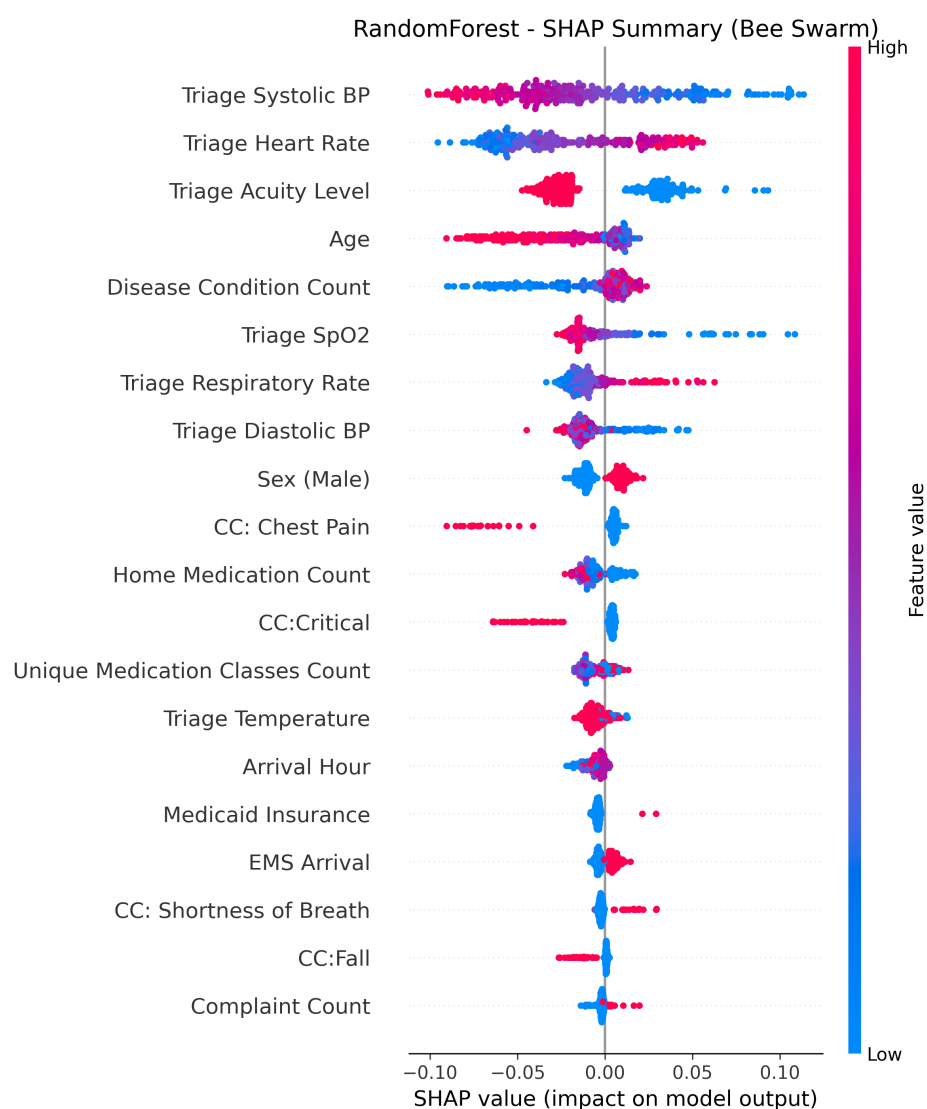


Figure 6. SHAP summary plot for Random Forest for hospital LOS >7 days prediction among admitted patients. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). Triage systolic blood pressure, triage heart rate, triage acuity level, age, and disease condition count are the top predictors.

procedure. Naemi et al. [28] reported that addressing missing data and skewness improved LOS prediction performance by an average of 15.75% in AUC, with classification-task AUC-ROC reaching approximately 0.82; our results are broadly consistent with the modest-to-moderate discrimination reported across this literature.

A key contribution of this study is the systematic SHAP analysis across all three prediction tasks [51, 52], which explains model behavior rather than establishing causal relationships. Triage acuity level, age, and comorbidity burden were consistently identified as the most important features across all three tasks. Beyond these common predictors, task-specific features emerged. EMS arrival was distinctively associated with prolonged ED length of stay in the model, consistent with the clinical expectation that ambulance-transported patients may have higher acuity and require more resources before disposition. Triage heart rate was among the top predictors for hospital admission, while vital signs (systolic blood pressure and heart rate) were important predictors of prolonged hospitalization. These SHAP-derived associations reflect patterns learned by the models and should be interpreted as hypothesis-generating rather than causal. The differences across tasks suggest that while baseline demographic and comorbidity

factors are universally important, physiological and logistical factors become more relevant at specific decision points in the model's predictions.

Subgroup sensitivity analyses revealed stable hospital admission performance across chief complaints (AUC range: 0.704–0.862) [48] and triage acuity levels. ED-LOS prediction showed declining discrimination with increasing age (AUC 0.688 in patients <65 years to 0.582 in patients ≥85 years), suggesting that the model performs less well in the oldest patients, who may have more complex clinical trajectories. Fairness analysis demonstrated relatively equitable performance across sex and race subgroups [36, 43].

Missing data were handled using median imputation for continuous vital signs, mode imputation for categorical variables (e.g., triage acuity), and listwise deletion for missing target values, based strictly on training set distributions to prevent data leakage [38, 42]. The absence of medication or diagnosis records in pmh.csv or meds.csv was treated as true zero (i.e., no documented medications or conditions), not as missing data.

Calibration analysis revealed that while the models rank patients well, the raw predicted probabilities require recalibration before clinical deployment. The calibration slopes deviating from 1.0 (ED-LOS XGBoost: slope 0.693) indicate that probability calibration via isotonic regression or Platt scaling would be necessary for use as absolute risk estimates.

Decision curve analysis provided complementary insights into clinical utility beyond discrimination metrics. For hospital admission, all four models demonstrated substantial net benefit across a wide threshold range (0.1–0.8), confirming their clinical utility for early disposition decisions. In contrast, ED-LOS and hospital LOS models showed more limited net benefit, primarily in the 0.2–0.4 threshold range, reflecting their more modest discriminative performance (AUC 0.660 and 0.670, respectively). These findings suggest that while ED-LOS and hospital LOS predictions may be useful for resource allocation and operational planning, their value for individual patient decision-making is limited without further improvement or integration with real-time operational data. The admission model, however, appears clinically actionable for individual disposition decisions.

Several important determinants of ED-LOS are not captured in our dataset, including inpatient boarding time, bed availability, staffing levels, laboratory turnaround time, and imaging delays. These operational factors are well-established drivers of ED crowding and prolonged length of stay [26, 27]. Our triage-time model necessarily cannot incorporate these downstream operational variables, which represents both a limitation and an opportunity: our predictions reflect the patient's intrinsic clinical complexity at arrival, while operational factors represent modifiable system-level contributors. Future work integrating real-time operational data (e.g., current ED census, inpatient bed availability) with our clinical predictions could yield even more accurate ED-LOS forecasts.

The results of this study could inform future prospective studies evaluating ED management optimization [44, 41]. By predicting prolonged ED-LOS, decision makers could potentially allocate additional medical and nursing staff during critical time slots. By predicting hospital admission, bed management could be improved. By predicting prolonged hospital LOS, discharge planning could be initiated earlier. However, these benefits require prospective validation.

4.1 Limitations

This study has several limitations. First, the single-centre retrospective design during the COVID-19 pandemic limits generalizability, and operational predictors such as bed availability and staffing were unavailable [62, 63, 26, 27]. Second, despite timestamp-based filtering, residual temporal leakage cannot be entirely ruled out. Third, the modest discrimination (ED-LOS AUC 0.660; Hosp-LOS AUC 0.670) and miscalibration observed limit standalone clinical utility, requiring probability recalibration and prospective validation before deployment [60, 61]. Fourth, sample sizes for some race subgroups (e.g., Black, n=591) limit fairness estimate precision, and prospective studies are needed to confirm real-world performance.

5 CONCLUSION

This study demonstrates that machine learning using data available at or before triage time can effectively predict hospital admission (AUC 0.816, strong discrimination) and modestly predict prolonged ED length of stay (AUC 0.660) and prolonged hospitalization (AUC 0.670) among admitted patients. Using the MC-MED dataset of 118,385 adult ED visits, Random Forest performed best for ED-LOS and hospital LOS prediction, while XGBoost achieved the best performance for hospital admission prediction.

SHAP analysis revealed that triage acuity level, age, and comorbidity burden were consistently important across all three tasks. Beyond these common predictors, EMS arrival was distinctively associated with prolonged ED length of stay, triage heart rate was strongly associated with admission decisions, and vital sign abnormalities were important predictors of prolonged hospitalization. These findings highlight how different clinical factors become relevant at different stages of patient care.

Decision curve analysis demonstrated strong clinical utility for hospital admission prediction across a wide threshold range (0.1–0.8) and more modest utility for ED-LOS and hospital LOS prediction, primarily in the 0.2–0.4 threshold range. Calibration analysis revealed systematic miscalibration across all tasks, indicating that probability recalibration is necessary before clinical deployment.

The admission model shows particular promise for early disposition support, while the ED-LOS and hospital LOS models provide complementary operational intelligence. However, all models require prospective validation, probability recalibration, and rigorous evaluation of clinical workflow integration before deployment [39]. These findings should be interpreted as hypothesis-generating for future prospective studies rather than confirming established clinical benefits. External validation across diverse healthcare settings is necessary, and models would require recalibration for new populations. This study provides an evidence base for future prospective evaluations of machine learning-based clinical decision support in emergency departments to improve patient flow and optimize resource allocation.

6 CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

7 AUTHOR CONTRIBUTIONS

Onyedikachi I. Onwurah designed the study, implemented all machine learning models, performed the data analysis, and drafted the manuscript. Sol Richardson contributed to study design, data interpretation, and critical revision. Wu Sheng served as clinical reviewer. Xia Tong supervised the research, guided the methodological approach, reviewed all versions of the manuscript, and approved the final submission.

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9 ETHICS STATEMENT

The Stanford University Institutional Review Board approved the original data collection for the MC-MED dataset with a waiver of informed consent [53]. This study involves secondary analysis of the publicly available, de-identified MC-MED dataset, which is exempt from additional IRB review under 45 CFR 46.104(d)(4) as research involving the secondary use of existing, de-identified data.

10 DATA AVAILABILITY STATEMENT

The MC-MED dataset is publicly available on PhysioNet at <https://physionet.org/content/mc-med/1.0.1/> [53]. Analysis code is available at <https://github.com/Drglazizzo/ED-visit-12ML-prediction>.

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12 PUBLISHER'S NOTE

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Table 2. Variables available at or before triage with derivation methods, N=29

Feature Name	Data Nature or Derivation
Demographics (7)	
Age	Direct from EHR, range 18–90 years (ages >90 capped at 90 per de-identification)
Sex (Male)	Direct from EHR, binary
Race (White)	Direct from EHR, binary
Race (Black)	Direct from EHR, binary
Hispanic Ethnicity	Direct from EHR, binary
Medicare Insurance	Direct from EHR, binary
Medicaid Insurance	Direct from EHR, binary
Triage Vitals and Acuity (7)	
Triage Temperature	Direct from EHR, degrees Celsius
Triage Heart Rate	Direct from EHR, beats per minute
Triage Respiratory Rate	Direct from EHR, breaths per minute
Triage SpO2	Direct from EHR, percentage
Triage Systolic Blood Pressure	Direct from EHR, mmHg
Triage Diastolic Blood Pressure	Direct from EHR, mmHg
Triage Acuity Level	Derived by extracting numeric prefix from strings “1-Resuscitation” through “5-Non-Urgent”, range 1 to 5
Arrival Context (3)	
Arrival Hour	Derived from Arrival_time timestamp, range 0 to 23
Weekend Arrival	Derived from Arrival_time, binary indicator for Saturday or Sunday
EMS Arrival	Direct from EHR, binary indicator for Means_of_arrival equal to “EMS”
Chief Complaint (7)	
Complaint Count	Derived as number of presenting complaints from the chief complaint string
CC: Abdominal Pain	Derived by keyword matching for “ABDOMINAL PAIN”
CC: Chest Pain	Derived by keyword matching for “CHEST PAIN”
CC: Shortness of Breath	Derived by keyword matching for “SHORTNESS OF BREATH”, “DYSPNEA”, “SOB”
CC: Fall	Derived by keyword matching for “FALL”, “FELL”, “FALLS”
CC: Fever	Derived by keyword matching for “FEVER”, “TEMP”, “HYPERTHERMIA”
CC: Critical	Derived by keyword matching for suicide, overdose, trauma, bleeding, seizure, stroke, cardiac arrest, respiratory failure
Comorbidity Burden (2)	
Charlson Comorbidity Index	Derived using the Quan et al. (2005) ICD-10 mapping algorithm from pre-arrival diagnoses, range 0 to 37
Disease Condition Count	Derived as unique Clinical Classifications Software categories per patient from pre-arrival ICD-10 codes
Home Medications (3)	
Home Medication Count	Derived as count of unique medication entries per MRN from meds.csv with Entry_date ≤ index Arrival_time
Unique Medication Classes Count	Derived as distinct Med_class categories per MRN from pre-arrival medications
Polypharmacy Flag	Derived as binary indicator for Home Medication Count ≥5

Table 3. Hyperparameter search spaces and optimal values for Bayesian optimization (task-specific shown)

Model	Hyperparameter	Search Range	Optimal Value	Task
XGBoost	max depth	3 to 10	6	All
	learning rate	0.01 to 0.3	0.05	All
	n estimators	100 to 500	500	All
	subsample	0.6 to 1.0	0.8	All
	colsample bytree	—	0.8	All
	scale pos weight	—	4.26	ED-LOS $\geq 8h$
	scale pos weight	—	1.49	Hospital Admission
	scale pos weight	—	5.04	Hosp-LOS $> 7d$
LightGBM	num leaves	10 to 100	31	All
	learning rate	0.01 to 0.3	0.05	All
	n estimators	100 to 500	500	All
	subsample	0.6 to 1.0	0.8	All
	colsample bytree	0.6 to 1.0	0.8	All
	reg alpha	—	0.01	All
	reg lambda	—	0.01	All
Random Forest	n estimators	100 to 500	500	All
	max depth	5 to 20	15	All
	min samples split	2 to 20	20	All
	min samples leaf	—	10	All
	max features	sqrt	sqrt	All
TabNet	n steps	1 to 5	5	All
	n d	8 to 64	64	All
	n a	8 to 64	64	All
	gamma	1.0 to 2.0	1.5	All
	lambda sparse	—	1e-4	All

Table 4. Evaluation metrics for the four ML algorithms on the test set with 95% confidence intervals

Task	Algorithm	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-score	AUC-ROC
ED-LOS ≥ 8 h	XGBoost	56.2	29.9	71.6	51.8	0.422	0.659 [0.648–0.670]
	LightGBM	54.1	28.5	75.4	49.8	0.416	0.648 [0.637–0.659]
	RF	56.0	29.8	65.1	55.3	0.422	0.660 [0.649–0.671]
	TabNet	55.3	29.2	60.2	56.4	0.411	0.646 [0.635–0.657]
Hospital Admission	XGBoost	67.0	63.9	62.6	70.6	0.632	0.816 [0.809–0.824]
	LightGBM	66.9	63.8	62.6	70.5	0.632	0.817 [0.810–0.825]
	RF	65.0	62.5	58.1	73.7	0.684	0.812 [0.804–0.820]
	TabNet	66.6	63.6	64.2	67.5	0.695	0.811 [0.803–0.819]
Hosp-LOS > 7 d*	XGBoost	57.6	26.0	69.8	53.3	0.387	0.655 [0.646–0.664]
	LightGBM	58.7	27.1	58.0	63.4	0.377	0.647 [0.638–0.656]
	RF	70.2	30.2	62.3	72.0	0.407	0.670 [0.662–0.680]
	TabNet	61.8	28.6	54.3	68.4	0.383	0.651 [0.642–0.660]

*Among admitted patients only (test set n=4,765). Metrics are reported at thresholds selected by maximizing Youden's J on the validation set (ED-LOS: 0.421; Admission: 0.398; Hosp-LOS: 0.412).

Table 5. Subgroup sensitivity analysis results across all three prediction tasks

Subgroup	AUC-ROC (95% CI)					
	ED-LOS ≥ 8 h		Hospital Admission		Hosp-LOS > 7 d	
Chief Complaint (not mutually exclusive)*						
Abdominal Pain	0.622 (n=1,619)	[0.601–0.643]	0.762 (n=1,619)	[0.741–0.783]	0.682 (n=581)	[0.653–0.711]
Chest Pain	0.679 (n=926)	[0.654–0.704]	0.807 (n=926)	[0.786–0.828]	0.661 (n=394)	[0.626–0.696]
Critical	0.657 (n=1,377)	[0.634–0.680]	0.804 (n=1,377)	[0.785–0.823]	0.667 (n=524)	[0.638–0.696]
Fall	0.606 (n=688)	[0.578–0.634]	0.779 (n=688)	[0.754–0.804]	0.664 (n=386)	[0.630–0.698]
Fever	0.643 (n=410)	[0.610–0.676]	0.862 (n=410)	[0.838–0.886]	0.629 (n=162)	[0.578–0.680]
Other/Unspecified	0.675 (n=7,466)	[0.666–0.684]	0.821 (n=7,466)	[0.813–0.829]	0.677 (n=2,883)	[0.665–0.689]
Shortness of Breath	0.615 (n=658)	[0.591–0.639]	0.824 (n=658)	[0.802–0.846]	0.686 (n=298)	[0.650–0.722]
Age Group						
<65 years	0.688 (n=8,058)	[0.677–0.699]	0.805 (n=8,058)	[0.795–0.815]	0.696 (n=2,336)	[0.683–0.709]
65–74 years	0.613 (n=1,739)	[0.594–0.632]	0.762 (n=1,739)	[0.743–0.781]	0.664 (n=948)	[0.641–0.687]
75–84 years	0.598 (n=1,374)	[0.574–0.622]	0.718 (n=1,374)	[0.695–0.741]	0.669 (n=902)	[0.645–0.693]
≥ 85 years	0.582 (n=849)	[0.553–0.611]	0.704 (n=849)	[0.677–0.731]	0.613 (n=579)	[0.584–0.642]
Triage Acuity						
Acuity 1 (Resuscitation)	0.541 (n=95)	[0.460–0.622]	0.770 (n=95)	[0.688–0.852]	0.693 (n=33)*	[0.573–0.813]
Acuity 2 (Emergent)	0.610 (n=3,180)	[0.595–0.625]	0.774 (n=3,180)	[0.761–0.787]	0.645 (n=1,347)	[0.626–0.664]
Acuity 3 (Urgent)	0.650 (n=7,510)	[0.640–0.660]	0.757 (n=7,510)	[0.747–0.767]	0.684 (n=3,019)	[0.672–0.696]
Acuity 4–5 (Semi/Non-urgent)	0.714 (n=1,235)	[0.691–0.737]	0.729 (n=1,235)	[0.703–0.755]	0.719 (n=366)	[0.685–0.753]

*Chief complaint subgroups are not mutually exclusive; patients with multiple complaints appear in each relevant subgroup.

*Small sample size (n=33) warrants cautious interpretation.